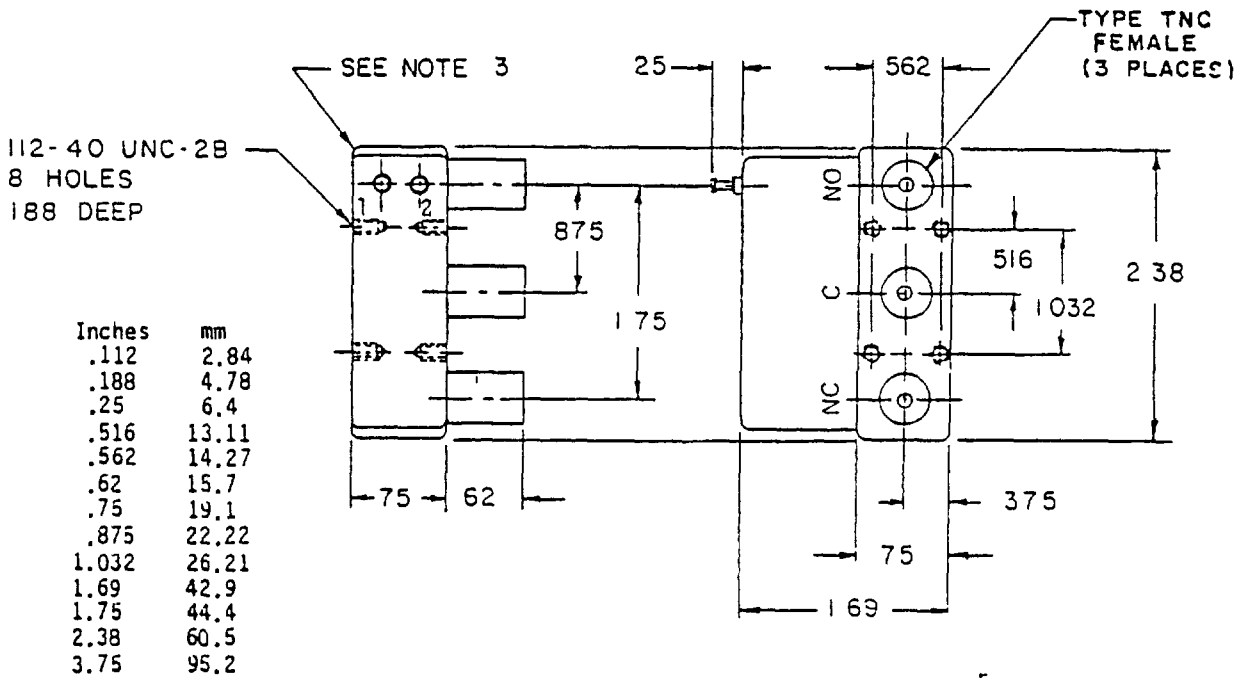
MILITARY SPECIFICATION SHEET

Ⓒ SWITCHES, RADIO FREQUENCY
TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED)
CLASS 4, 1P2T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.



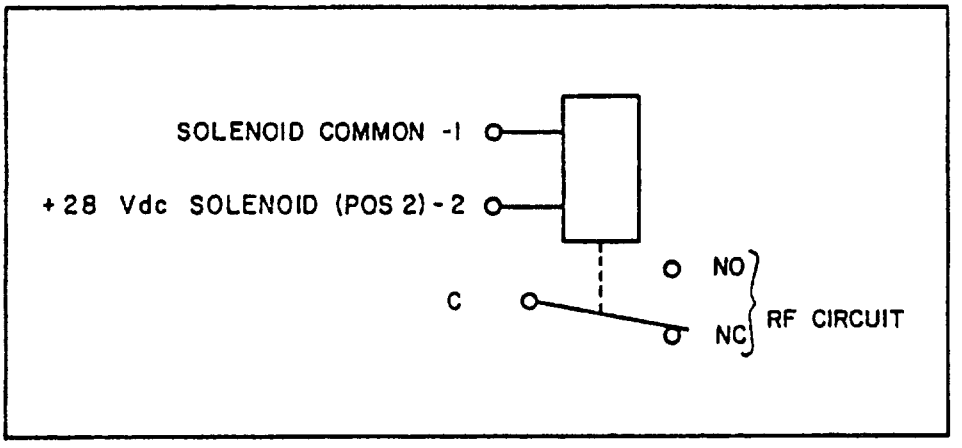
Inches	mm
.112	2.84
.188	4.78
.25	6.4
.516	13.11
.562	14.27
.62	15.7
.75	19.1
.875	22.22
1.032	26.21
1.69	42.9
1.75	44.4
2.38	60.5
3.75	95.2

Ⓒ FIGURE 1. Part number M3928/20-02.

Ⓒ denotes changes

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Position 1. (Deenergized or failsafe position),
C connects to NC connector.
NO is open circuit.

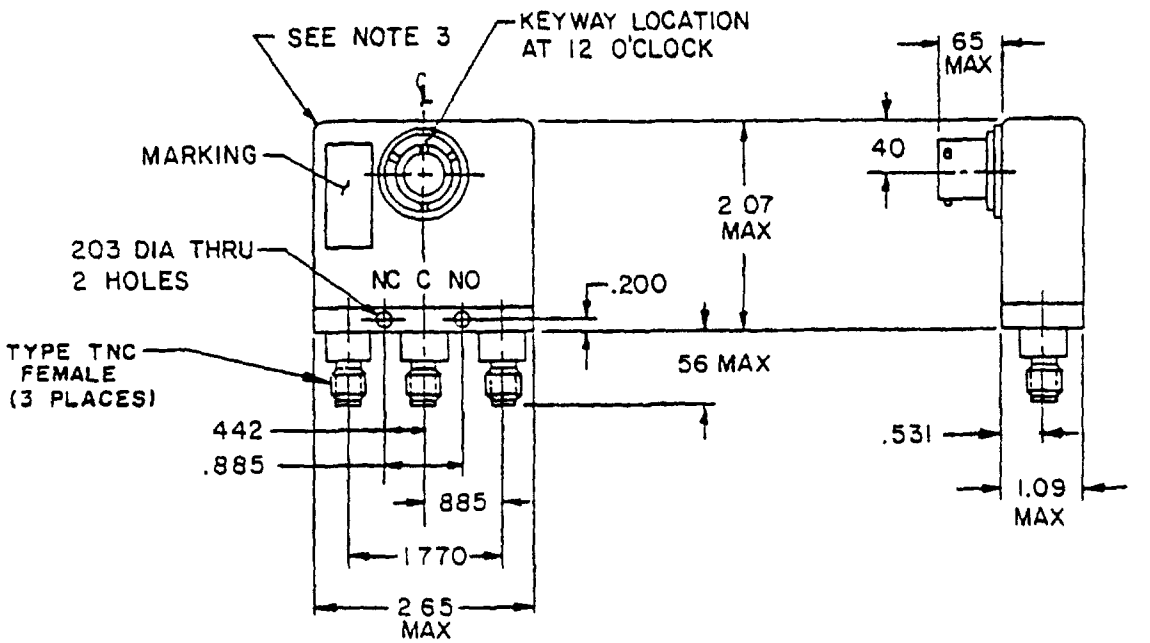
Position 2. (Energized position),
C connects to NO connector.
NC is open circuit.

SCHEMATIC (SHOWN IN POSITION 1)

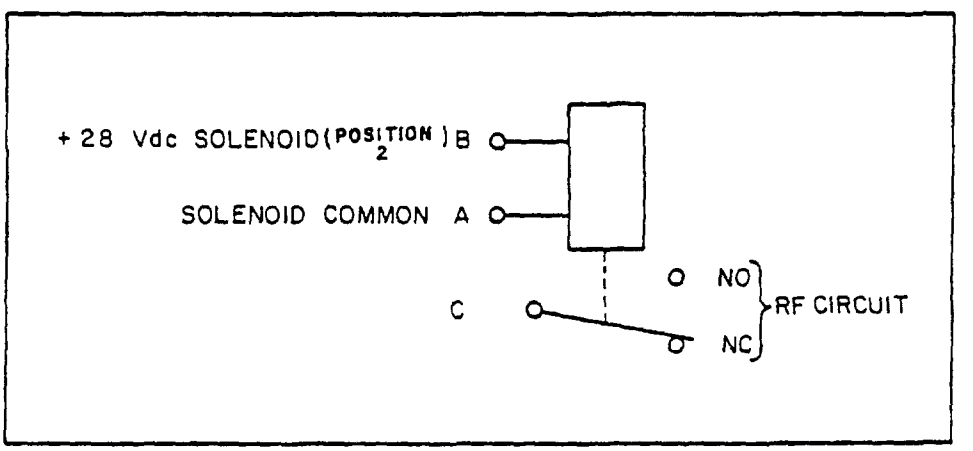
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 1. Part number M3928/20-02 - Continued.

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Inches	mm
.200	5.08
.203	5.16
.40	10.2
.442	11.23
.531	13.49
.56	14.2
.65	16.5
.870	22.10
.885	22.48
1 09	27.7
1.770	44.96
2.07	52.6
2.65	67.3



POSITION 1. (DEENERGIZED OR FAILSAFE POSITION) C CONNECTS TO NC CONNECTOR NO IS OPEN CIRCUIT.

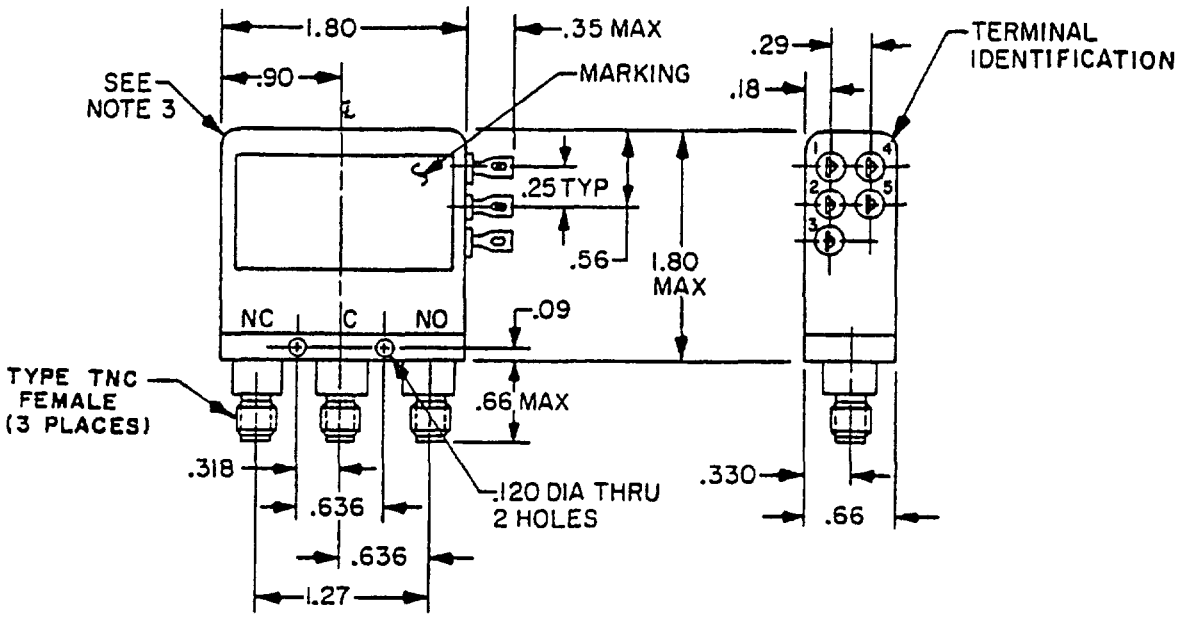
POSITION 2 (ENERGIZED POSITION) C CONNECTS TO NO CONNECTOR NC IS OPEN CIRCUIT

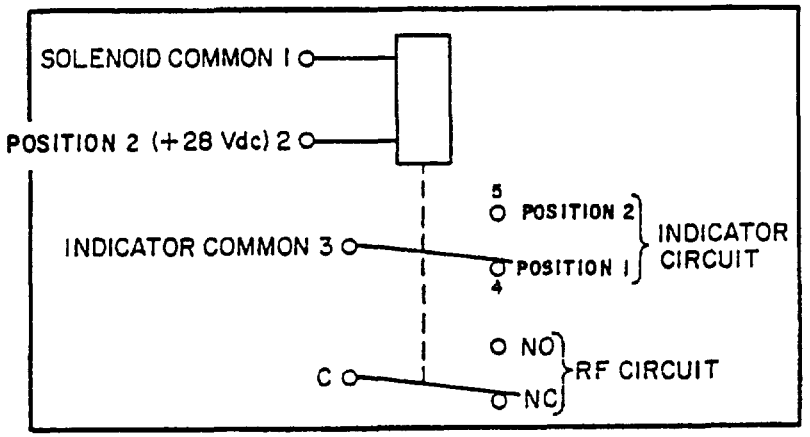
SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 2. Part number M3928/20-03.

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Inches	mm
.09	2.3
.120	3.05
.18	4.6
.25	6.4
.29	7.4
.318	8.08
.330	8.38
.35	8.9
.56	14.2
.636	16.15
.66	16.8
.90	22.9
1.27	32.3
1.80	45.7

POSITION 1. (DEENERGIZED OR FAILSAFE POSITION.) C CONNECTS TO NC CONNECTOR. NO IS OPEN CIRCUIT.

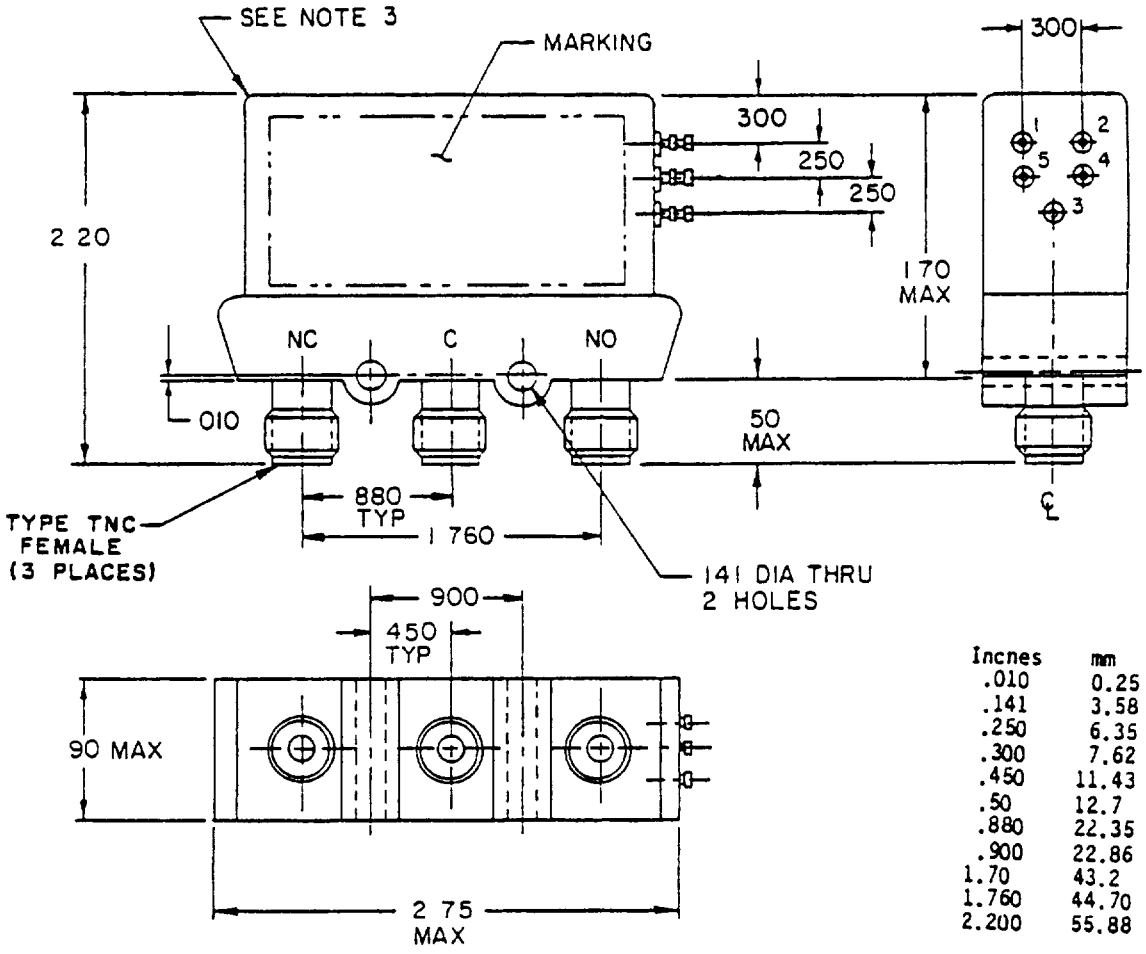
POSITION 2. (ENERGIZED POSITION.) C CONNECTS TO NO CONNECTOR. NC IS OPEN CIRCUIT.

SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 3. Part number M3928/20-04.

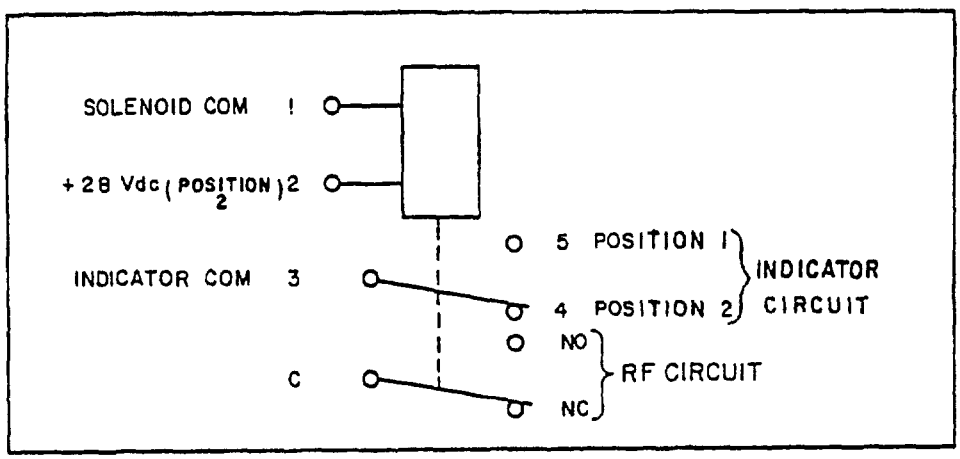
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T22:20Z
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Incnes	mm
.010	0.25
.141	3.58
.250	6.35
.300	7.62
.450	11.43
.50	12.7
.880	22.35
.900	22.86
1.70	43.2
1.760	44.70
2.200	55.88

© FIGURE 4. Part number M3928/20-05.

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POSITION 1 (DEENERGIZED OR FAILSAFE POSITION) C CONNECTS TO NC CONNECTOR NO IS OPEN CIRCUIT

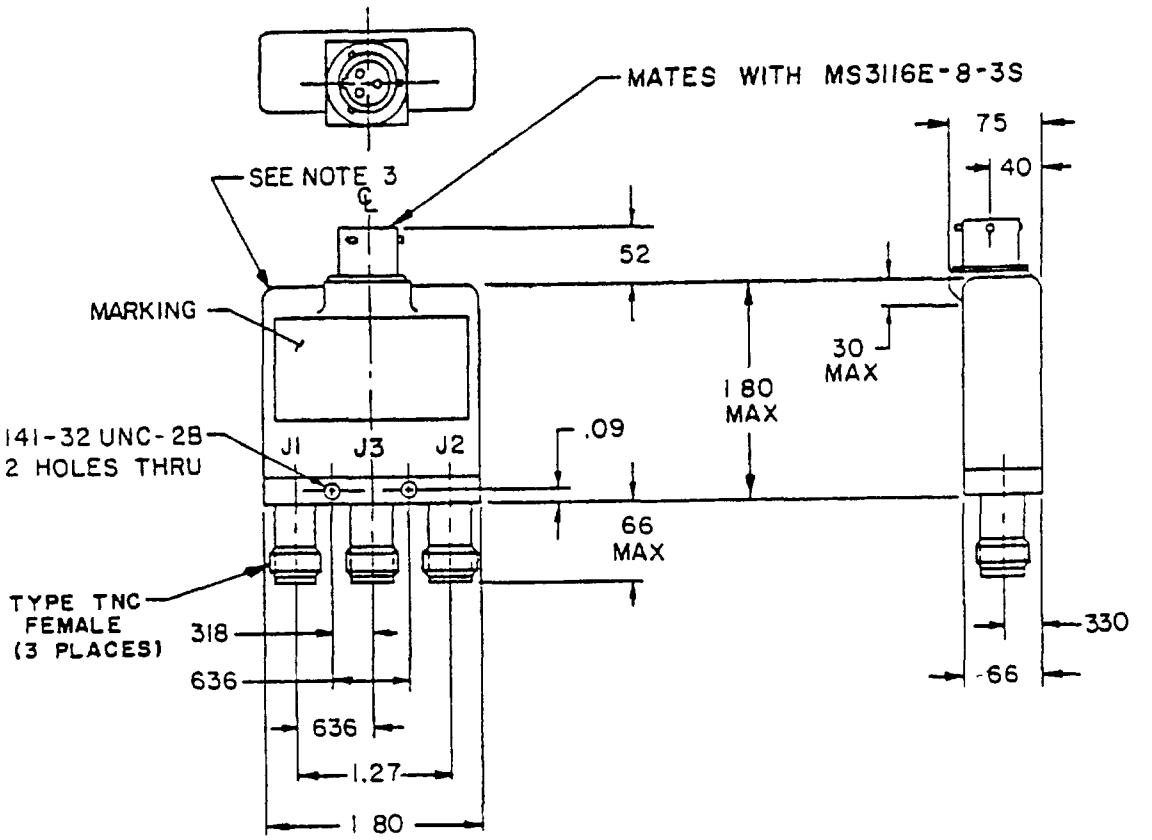
POSITION 2 (ENERGIZED POSITION.) C CONNECTS TO NO CONNECTOR NC IS OPEN CIRCUIT

SCHEMATIC (SHOWN IN POSITION 1)

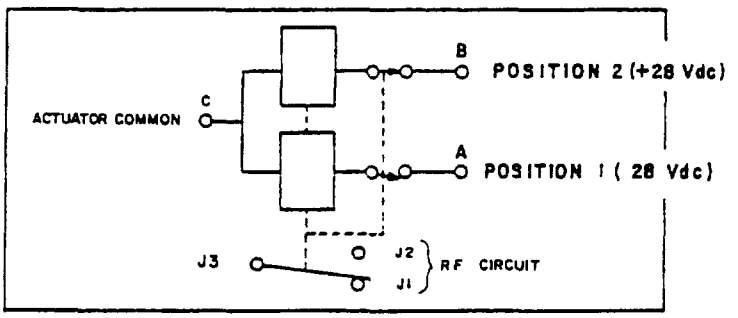
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 4. Part number M3928/20-05 - Continued.

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Inches	mm
.09	2.3
.141	3.58
.30	7.6
.318	8.08
.40	10.2
.52	13.2
.636	16.15
.66	16.8
.75	19.0
1.27	32.3
1.80	45.7



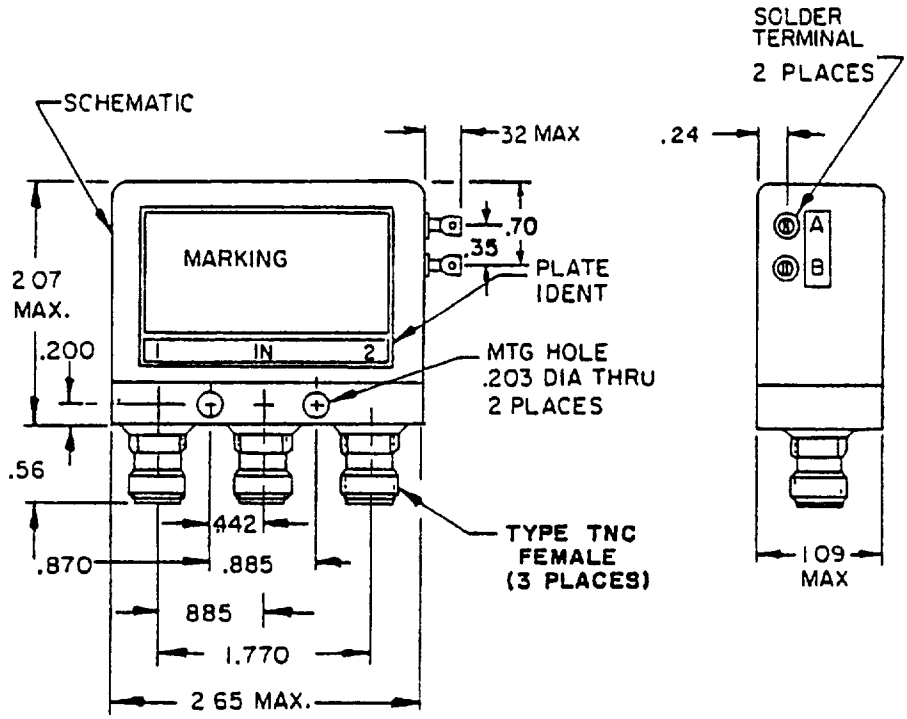
POSITION 1 ENERGIZE PIN A J3 CONNECTS TO J1, J2 IS OPEN CIRCUIT
POSITION 2 ENERGIZE PIN B J3 CONNECTS TO J2, J1 IS OPEN CIRCUIT

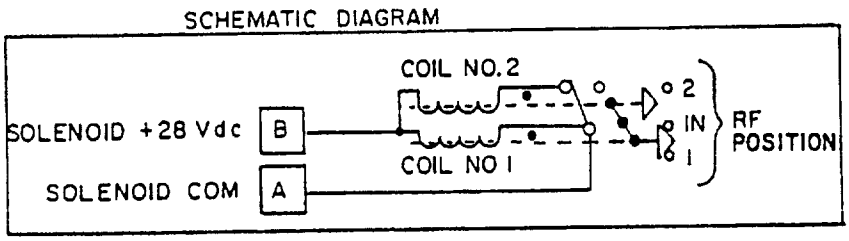
SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 5. Part number M3928/20-06

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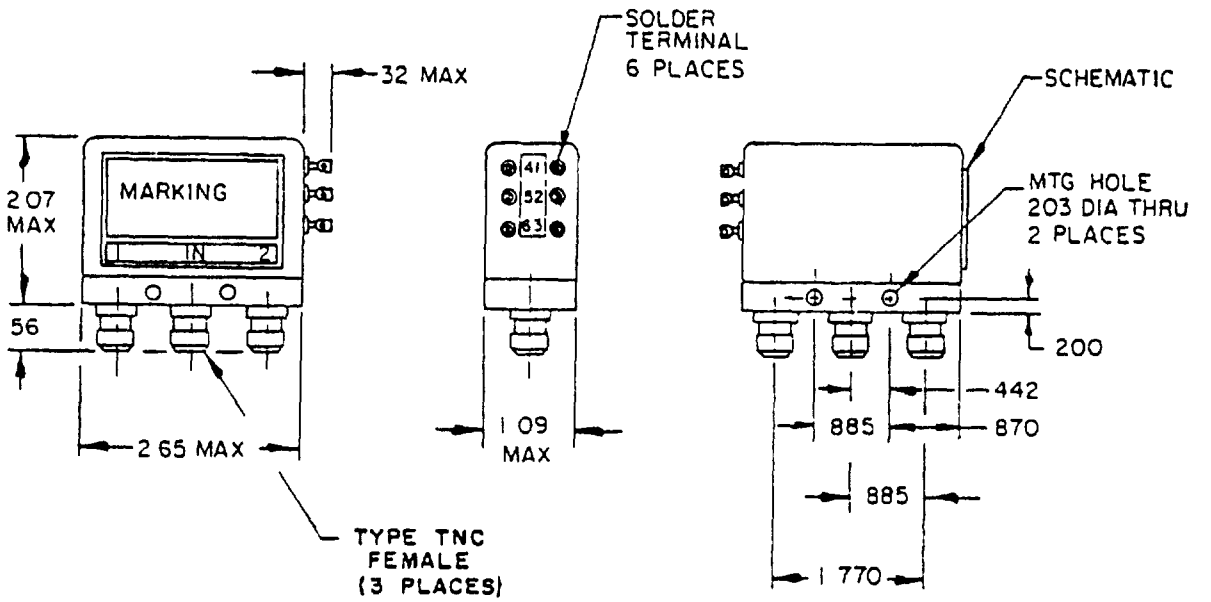
SHOWN IN FAILSAFE POSITION (POSITION 1)
ENERGIZED TERM B TO CONNECT IN TO 2 (POSITION 2)

Inches	mm
.200	5.08
.203	5.16
.32	8.1
.35	8.9
.442	11.23
.60	15.2
.70	17.8
.870	22.10
.885	22.48
1.09	27.7
1.770	44.96
2.07	52.6
2.65	67.3

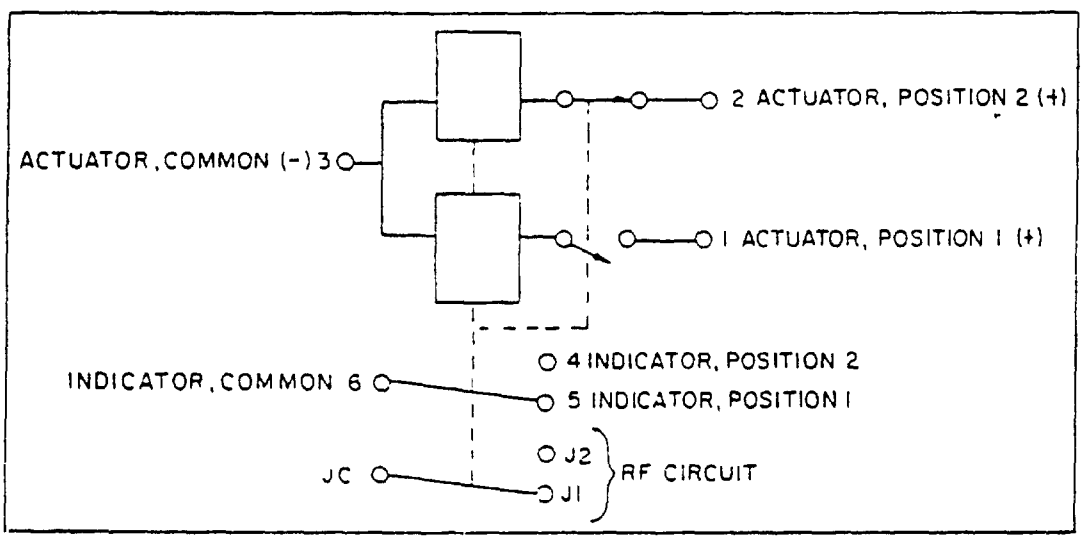
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 6. Part number M3928/20-07.

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Inches	mm
.200	5.08
.203	5.16
.24	6.1
.32	8.1
.442	11.23
.60	15.2
.870	22.10
.885	22.48
1.09	27.7
1 770	44.96
2.07	52.6
2.65	67.3



POSITION 1. ENERGIZE TERMINAL 1. JC CONNECTS TO J1, J2 IS OPEN CIRCUIT.

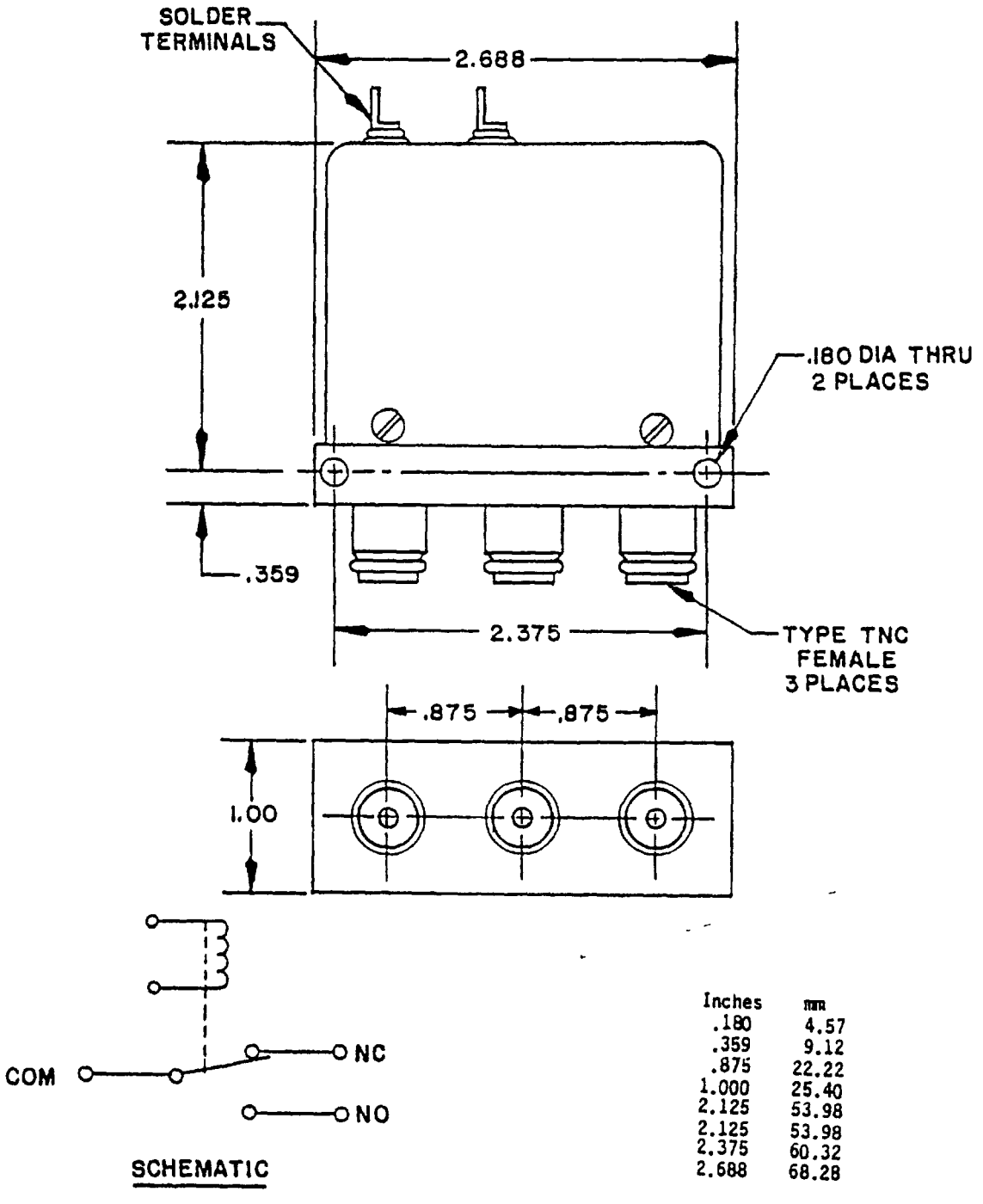
POSITION 2. ENERGIZE TERMINAL 2. JC CONNECTS TO J2, J1 IS OPEN CIRCUIT.

SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 7. Part number 43928/20-08.

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SCHEMATIC

Inches	mm
.180	4.57
.359	9.12
.875	22.22
1.000	25.40
2.125	53.98
2.125	53.98
2.375	60.32
2.688	68.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

© FIGURE 8. Switch configuration and schematic, part number M3928/20-09.

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(f) TABLE I Electrical and performance characteristics 1/

Part no M3928/2( )	Figure no	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switch time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail safe or latching	Operating current (A) 2/	Holding current (A) 2/	Nominal operating voltage	Pickup voltage	Dropout voltage	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
02N S	2	E	6	1 5 1	0 3	50	25	None	1000	F	0 12	0 12	28 V dc	20 V dc	5 V dc	Solder terminals	4
03N,S	3	E	11	1 4 1	0 4	60	20	None	100	F	0 8	0 4	28 V dc	18 V dc	5 V dc	MS3113-8-2P	10
3/ 04N S	4	E	11 25	1 4 1	0 5	50	20	0 2A @ 40 V dc	100	F	0 28	0 28	28 V dc	22 V dc	5 V dc	Solder terminals	4
05N S	5	E	0 4	1 2 1	0 15	60	25	2A @ 28 V dc	1000	F	0 25	0 25	28 V dc	20 V dc	5 V dc	Solder terminals	6
3/ 06N S 4/	6	E	12 4	1 5 1	0 5	50	20	None	1000	L	0 56	0	28 V dc	24 V dc	---	Mates with MS3116E 8 3S	4
07N S	7	E	12 4	1 5 1	0 5	60	20	None	100	F	0 80	0 40	28 V dc	20 V dc	10 V dc	Solder terminals	9 5
3/ 08N,S	8	E	12 4	1 5 1	0 5	50	20	0 1A @ 28 V dc	100	L	0 56	---	28 V dc	20 V dc	---	Solder terminals	4
5/			@1 57	1 2 1	0 15	60	30										
09N S	9	E	12 4	1 5 1	50	60	15	None	1000	F	0 12	0 12	28 V dc	20 V dc	8 V dc	Solder terminals	7

1/ M3928/20 01 has been deleted

2/ At 28 V dc and 20 L

3/ Transient interference (RFI)

4/ Sequence make before break

5/ RF power handling capacity 80 watts minimum at sea level and 10^{5} through 10^{-10} torr

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REQUIREMENTS:

Ⓒ Dimensions and configurations: See figures 1 through 8.

RF connectors: Female connectors (3 places) shall meet the requirements of MIL-C-39012 and shall mate with TNC type male connectors per MIL-C-39012/26.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 50 watts (minimum).

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part number: M3928/20- (dash number from table I).

Substitutability: Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table II. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

Ⓒ TABLE II. Substitution data.

Part number M3928/18-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985
02	81413	224534-000-001	00-113-4940
02	74868	360-11892-15	00-113-4940
03	81205	37-1021-01	00-417-0532
03	82152	810C30900	00-417-0532
04	07397	20-811076-1	00-006-4308
04	82152	910C90700	00-006-4308
05	28528	1131510GL	00-630-6634
05	24022	CS31T1C-1A	---
06	78022	116E500-1	00-619-7145
06	82152	900C31500	00-619-7145
07	05395	S30139-1	---
07	82152	810C30100	00-248-2974
07	96067	MA7524PTD	---
07	04660	72C861-001	00-248-2974
08	28528	1133649G1	---
08	82152	805C30200	---
08	24022	CS-31T6B	---
09	31804	20A0243003	---
09	74868	303-10168-3	---

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent.
DLA - ES

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